Design, Development, and Flight Testing of a High Endurance Micro Quadrotor Helicopter

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Abstract

This paper describes the experimental design studies conducted to maximize the hover endurance of a MAV-scale quadrotor helicopter by careful design/choice of the individual subsystems while understanding their interdependencies. The different subsystems of the quadrotor that were examined include the rotors, motors, electronic speed controllers, gear transmission systems, battery, and airframe structure. An optimized MAV-scale rotor with a diameter of 110 mm and a figure of merit of 0.67 served as the basis for the full vehicle design. Brushed and brushless motor systems were paired with the optimized rotor and were examined for maximum system level efficiency and capability to produce the required thrust. Propulsion systems using brushed motors with speed controllers and the optimum gear ratio showed up to 35% improvement in efficiency over brushless outrunner motors with speed controllers. Lithium-ion polymer batteries were compared and evaluated on voltage drop-off characteristics and achievable endurance vs. weight. Quadrotor airframe weight has been minimized to 7.4% of the gross weight, representing a 30 - 50% decrease compared existing similarly sized quadrotors. The highest endurance was obtained using brushed motors with a 5.33:1 gear ratio and a 650mAh battery. Presently, the longest continuous hover endurance achieved with this design is 31 minutes.

Introduction

Micro Air Vehicles (MAVs) is a rapidly emerging field of research, which is envisioned to have a wide range of military and civilian applications. They offer several advantages such as portability, rapid deployment, real-time data acquisition capability, low radar cross section, low noise signatures and low production cost. However, even though the concept of MAV appears attractive, the MAV research is still in its incipient stages. It should be noted that only a decade of research has gone into these small vehicles and the key technical barriers are only being resolved now. Some of these barriers include efficient small-scale power generation and storage, out-of-sight navigation and communications, low Reynolds number aerodynamics, and autonomous control.

Despite the advances made in quadrotor and multicopter dynamics and control within the past decade, they still exhibit very poor flight endurance compared to larger aircraft. This is evident in Table 1 which provides examples of typical rotary-wing MAVs developed by universities and commercial vendors with their associated flight endurance [1]-[5]. It should be noted that none of these rotary-wing MAVs can achieve even 15 minute flight endurance, meaning their mission capabilities are still fairly limited. Even the best performing vehicle, a quadrotor developed at the University of Pennsylvania GRASP Lab, only has a hover endurance of 11 minutes [5]. Low endurance can be attributed to multiple factors such as motor efficiency and limited battery storage. Additionally, the small scale of these vehicles also means that they operate within a low Reynolds number range (10,000 – 50,000), which introduces further performance issues. In the MAV Reynolds number range, viscous forces dominate inertial forces. This results in lower-lift-to drag ratios, induced losses in the viscous-dominated rotor wake structure, and large laminar separation bubbles [6].

Most of the previous studies on MAV-scale rotors either involved comprehensive performance measurements to optimize efficiency [7],[8] or used CFD models or flowfield measurements to investigate the aerodynamics of one or two specific rotor designs [6],[9]. However, in a previous study, through comprehensive performance measurements by varying different rotor parameters (such as blade airfoil section, blade chord, number of blades, blade twist and planform) along with high-fidelity flowfield measurements (using Particle Image Velocimetry, PIV), a micro-rotor optimized for low Reynolds number (RE<30,000) operation has been developed [10].

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Table 1. Typical Rotary-Wing MAV Performance [1]-[5]

MAV	Largest Dimension, cm	Weight, g	Endurance, min
Heli-Max 1Si (Quadrotor)	13.8	45.9	5
Scorpion Mini Multicopter RTF (Multicopter)	12	55	6.5
MICOR (Coaxial)	15.24	155	10
QR W100S (Quadrotor)	14.5	89	10
GRASP Micro Quadrotor	21	73	11

Even though improving the aerodynamic efficiency of the rotor is a crucial step, increasing MAV hover endurance will require maximizing the overall system efficiency. Current research-based and commercial micro quadrotors are not designed to efficiently extract the maximum flight endurance from a limited power source for multiple reasons [11]. The rotors used have not been designed for low Reynolds number application. The motors and transmissions have not been appropriately paired with the rotors for maximum efficiency. Current designs where brushless outrunner motors directly drives the rotor simplifies the mechanical design, however, a huge price is paid in terms of motor efficiency. Also motor speed controller losses, particularly for brushless motors is much higher than that of brushed motors. Battery performance characteristics have not been incorporated into the design [11]. And, the airframe structure is heavily overdesigned and constitutes a large fraction of the gross vehicle weight. The present research aims to examine each of these factors and their interactions to improve system level efficiency.

This paper details the systematic performance studies as follows: optimizing the micro-rotor performance through systematic experimental studies, evaluation of brushed and brushless motor system characteristics, optimization of the geared transmission system, evaluation of battery performance characteristics, implementation of control and stability hardware and algorithm, minimization of airframe weight, gimbal endurance test, and final free hover endurance results. The ultimate goal of these optimization studies is to develop an efficient micro quad-rotor weighing under 50 grams with a hover endurance over 30 minutes.

Rotor Design Optimization

The initial step in designing the micro quadrotor was choosing an optimized rotor for maximum figure of merit (*FM*). A comprehensive parametric study has been previously conducted by systematically varying multiple rotor characteristics [10]. Parameters that were examined include: blade airfoil section, blade collective pitch, camber, thickness, rotor solidity, number of blades, geometric twist, and planform shape. The baseline rotor had a diameter of 3.2 inches with a uniform chord of 0.43 inches. The rotor parameters were then factored into over 500 rotor designs that were fabricated, and tested through an iterative improvement process to arrive at the optimal rotor design [10]. The results of the micro-rotor optimization study are shown in Figures 1-5.

The performance results of varying blade airfoil section are represented by *FM* in Figure 1. From these results, it was observed that the three highest performing airfoils were the NACA 6504 (*FM* = 0.57), Eppler-63 (*FM* = 0.57), and 6.1% cambered plate (*FM* = 0.59) [10]. This was dramatic increase from the baseline rotor with a NACA 0012 airfoil (*FM* = 0.44). This large performance gap can be attributed to key airfoil characteristics: primarily camber and thickness-to-chord ratio (*t/c*). The NACA 0012 is symmetric and relatively thick with *t/c* = 12%, whereas the high performance airfoils are thin and moderately cambered. The cambers of these airfoils are: 6% (NACA 6504), 5.3% (Eppler-63), and 6.1% (cambered plate). Max camber position is at 50% chord for all three airfoils. The *t/c* of these airfoils are: 4% (NACA 6504), 4.3% (Eppler-63), and 2.2% (cambered plate) [10]. The key trend observed from these experiments was that low Reynolds number rotor efficiency increased as airfoil *t/c* is decreased to minimal structural limits, and airfoil camber is increased to approximately 6%.

The effect of airfoil thickness on rotor performance was further studied with incremental variations in NACA airfoil *t/c*. The thickness-to-chord ratios examined were 4%, 6%, 8% and 12%. This range of *t/c* was compared in sets of airfoils with constant cambers of 3%, 4%, 5%, and 6%. The *FM* performance results of varying *t/c* for 6% camber are shown in Figure 2. It is evident from these results that for 6% camber, rotor *FM* increases as airfoil *t/c* decreases [10]. Furthermore, similar results were observed in the 3%, 5%, and 6% camber sets. This agrees with the trend that emerged from the preliminary airfoil experiments. This indicates that

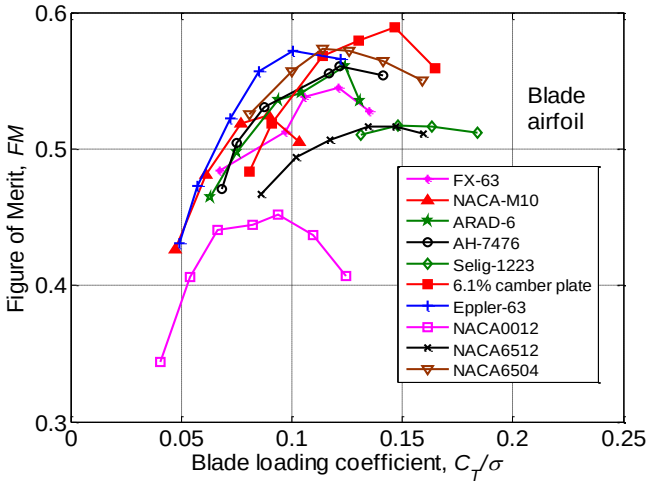


Figure 1. Performance of Rotors with Different Airfoil Sections in Terms of Figure of Merit [10]

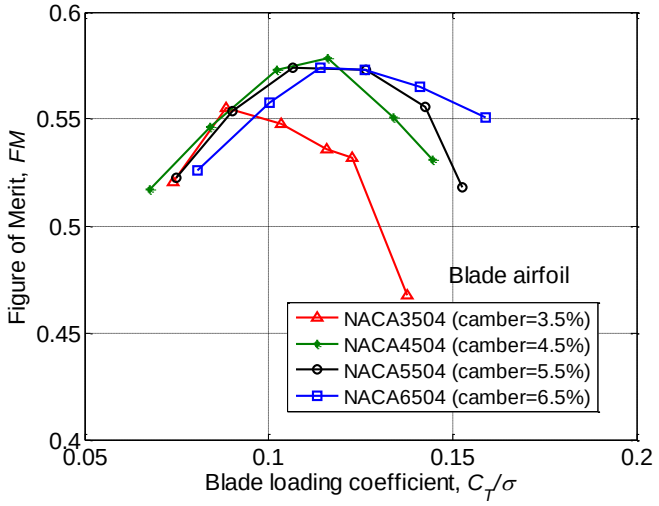


Figure 2. Effect of Airfoil Camber on Figure of Merit [10]

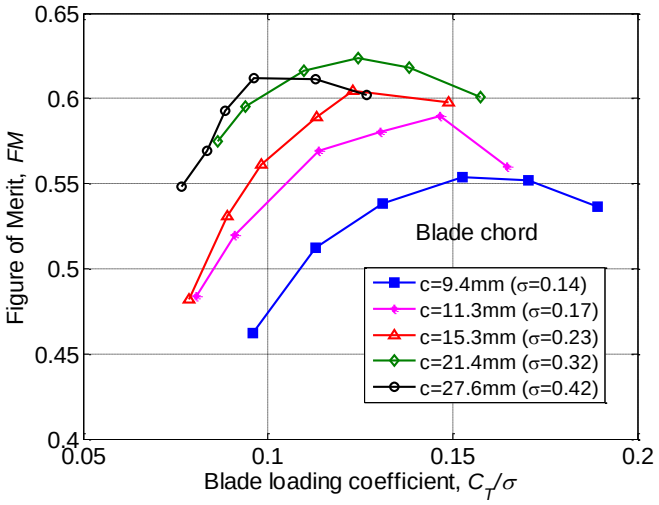


Figure 4. Effect of Increasing Chord on Figure of Merit [10]

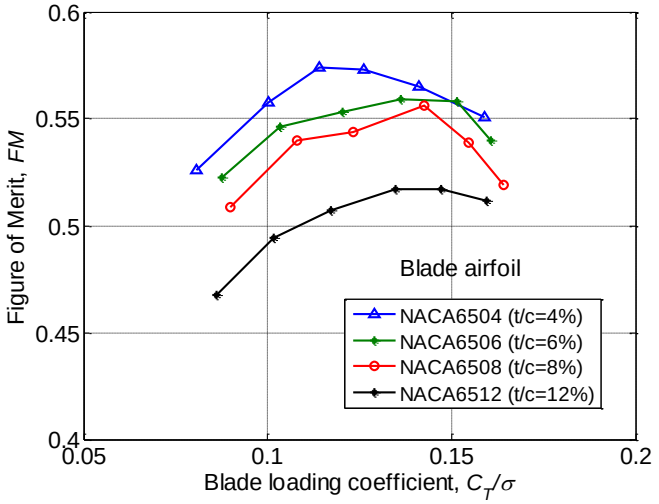


Figure 3. Effect of Airfoil Thickness on Figure of Merit [10]

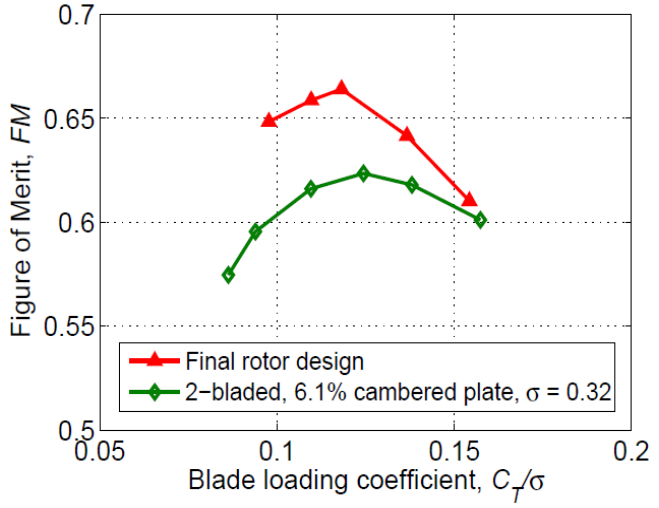


Figure 5. Final Rotor Design with Maximum Figure of Merit Achieved [10]

a crucial objective for optimizing low Reynolds number rotors is to minimize blade thickness [10].

Standard 4-digit NACA airfoils were used to further investigate the effect of camber on rotor performance. NACA airfoil cambers examined were 3%, 4%, 5%, and 6%. This range of camber was tested in sets of airfoils with constant *t/c* of 4%, 6%, 8%, and 12%. The *FM* performance results of varying camber for 4% *t/c* are shown in Figure 3. It is observed in these results that for both thickness-to-chord ratios, maximum *FM* is relatively unchanged for rotors with airfoil camber between 4% and 6% [10]. However, just below this range, at 3% camber, rotor performance drops significantly. This same trend was also observed for the 6%, 8%, and 12% *t*/c NACA airfoil sets. The results of these incremental-camber-variation tests indicate that camber values within the 4 – 6% range increased performance for micro-rotors [10].

After determining optimal rotor blade airfoil sections, the next major parameter investigated was rotor solidity (σ). Solidity tests were conducted by varying chord length and/or the number of blades. In all previous tests, rotors were 2 bladed with c = 11.3 mm and σ = 0.17. For these experiments, the following solidities and their corresponding chord lengths were investigated: σ = 0.14 (c = 9.4 mm), σ = 0.17 (c = 11.3 mm), σ = 0.23 (c = 15.3 mm), σ = 0.32 (c = 21.4 mm), and σ = 0.42 (c = 27.6 mm). The results of these experiments are shown in Figure 4 as *FM* performance curves. From these results, it is evident that performance for 2-bladed, 6.1% cambered plate rotors increases as chord length increases up to 21.4 mm [10]. It should be noted that the maximum *FM* of the high solidity (σ = 0.32) rotor greatly improved compared to the baseline solidity (σ = 0.17) rotor. Optimizing the chord length of the 2-bladed, 6.1% cambered plate rotor increased the maximum *FM* from 0.59 to 0.62 [10]. However, increasing solidity by increasing the number of blades, rather than chord length, did not produce significant performance gains.

Additional tests were also carried out to examine the effect of rotor twist and taper. Though only relatively small

performance gains could be achieved, results showed that rotor efficiency could further be optimized by incorporating a 0.25 to 0.5 chord taper ratio and 10° to 20° of negative twist [10].

The key observations from this previous study are that micro-rotor efficiency is most influenced by blade airfoil camber, thickness-to-chord (*t/c*) ratio, and chord length. The performance results of the final micro-rotor design are represented in Figure 5. The highest figure of merit achieved with this micro-rotor is 0.67, which represents a 34% increase in performance from the initial, baseline rotor [10]. The optimal rotor was fabricated from carbon fiber composite and has the following parameters: 2 blades, 0.32 thrust weighted solidity, 21.4mm chord at 75% span, 0.25mm thickness, 5-10% camber, 19.5° of collective blade pitch, -11.4° of twist, and a 0.5 chord taper ratio [10].

Electric Motor and Speed Controller Experiments

With the optimal rotor design chosen, further systematic tests were required to determine the optimal rotor-motor pairing in terms of motor efficiency:

$$Efficiency = \frac{Power\ Out}{Power\ In} = \frac{Torque * Rotational\ Speed}{Voltage * Current}$$

and electrical power loading (thrust produced per electrical power consumed). Both brushed and brushless motors were considered.

Brushed motors use an internal electrical commutator (brushes) that reverse the polarity of the active wire coils as they pass permanent magnets. Speed can be controlled simply by varying the input DC voltage. This is an advantage over brushless motors which require a three phase speed controller. Another advantage of brushed motors is that they have a low cost of construction. Furthermore, they can be easily miniaturized to even less than 0.5 gram [12].

The other main class of motors is the DC brushless motor. These motors eliminate the need for a commutator and brushes with the use of a three phase speed controller. These speed controllers take a DC power source input as well as a pulse width modulated (PWM) signal and in turn, electrically phase the voltage and current going to each winding phase of the motor accordingly to regulate speed [12]. This results in different advantages and disadvantages for brushless motors. Efficiency can be increased as there is no longer a voltage drop across the brushes and during commutation. Without brushes, internal friction is reduced and higher speeds can be attained. The main disadvantage of brushless motors is the higher cost of control, as a separate electronic speed controller (ESC) with much more complex electronics is needed [12]. These speed controllers are usually heavy and incur an additional weight penalty that is disadvantageous for micro air vehicle applications [12]. They also add a secondary source of inefficiency which will be discussed later.

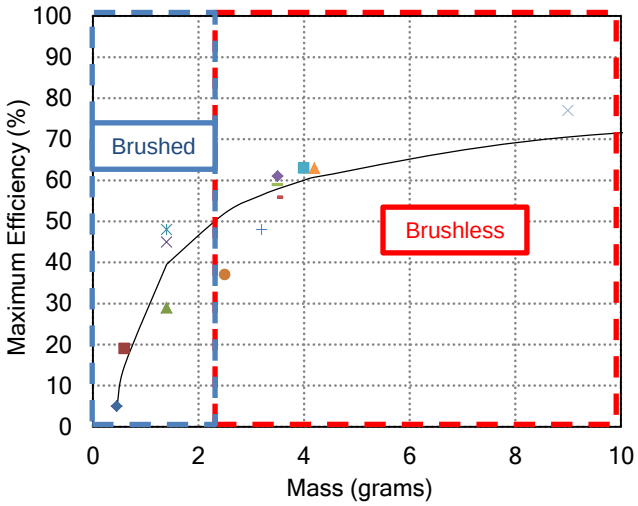


Figure 6. Maximum Efficiency vs Mass for a Representative Sample of Brushed and Brushless Motors [12]

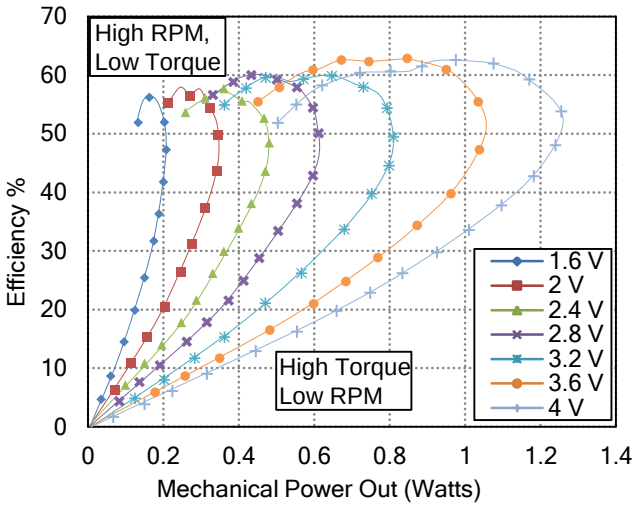


Figure 7. Efficiency vs. Motor Power Output for the SS-1.7 Brushed Motor [12]

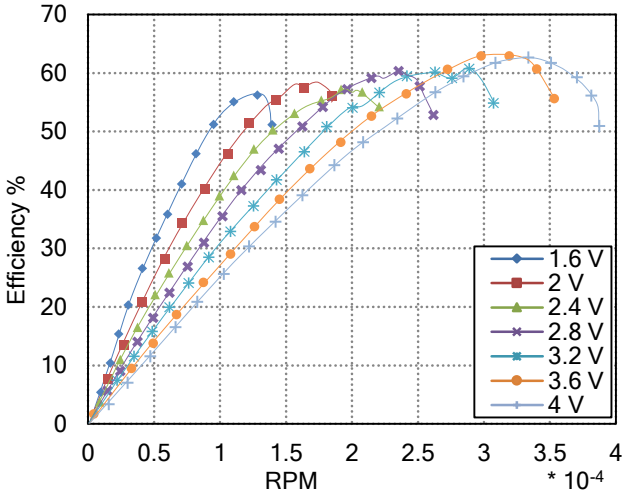


Figure 8. Efficiency vs RPM for SS-1.7 Brushed Motor [12]

Extensive performance studies on small (<5g) brushed and brushless motors have already been conducted by Harrington and Kroninger [12]. The major findings from these studies are summarized in the current section and were used as a starting point for systematically determining an optimal motor for a micro quadrotor. Figure 6 shows that, in general, motor efficiency increases with motor mass. However, beyond 5 grams, the efficiency gains are marginal as the mass becomes prohibitive for application on a micro quadrotor of size discussed in the present study. According to Figure 6, while the brushed motors are lighter, they also tend to be less efficient [12]. Conversely, brushless motors tend to be more efficient but heavier. This means there are only a few brushed motors with high enough efficiency and few brushless motors that are light enough to be considered for a 50g quadrotor design.

The motor efficiency-weight trade-off needed to be examined further to determine if the increased efficiency of brushless motors was enough to overcome the added weight. This requires an understanding of the operating conditions (torque and RPM) which effect motor efficiency. As seen in Figure 7 and Figure 8, brushed motor efficiency is strongly influenced by RPM and applied torque for a constant voltage. It is important to note from Figure 7 that there are two efficiency values for each power value because of the two possible combinations of RPM and torque. The "no-load side" of the curve is characterized by low torque and high RPM while the "stall side" is characterized by low RPM and high torque [12]. It is evident from Figure 7 and Figure 8 that the SS-1.7 brushed motor, pictured in Figure 9, is most efficient in high-RPM, low-torque applications. Furthermore, this same characteristic was observed all brushed motors, including the SS-2.3 and SS-3.3, in the Harrington, Kroninger study [12]. The SS-1.7, 2.3, and 3.3 (where the number designates the average internal resistance of the motor in Ohms) yielded the highest overall brushed motor performance in the study, and thus were selected for further optimization testing.

Brushless motors were shown to require different operating parameters compared to brushed motors for maximum efficiency. While both types of motors operate most efficiently at high speeds, brushless motors also require a higher torque to operate more efficiently [12]. Figure 10 is a representation of this characteristic for the AP03-4000 brushless motor. It shows brushless motor efficiency as a function of torque and RPM. It is important to note that peak efficiency for the AP03-4000 occurs at a large value of torque (1.5mN-m) and high rotational speed (1.75×10^4 RPM). Furthermore, this trend was generally observed in all other brushless motors in the Harrington, Kroninger study [12]. The AP03-4000 and AP03-7000 were concluded to have the best brushless performance characteristics while still being lightweight (<4g). For these reasons, the performance of the AP03-4000 and AP03-7000 were evaluated with the addition of a speed controller.

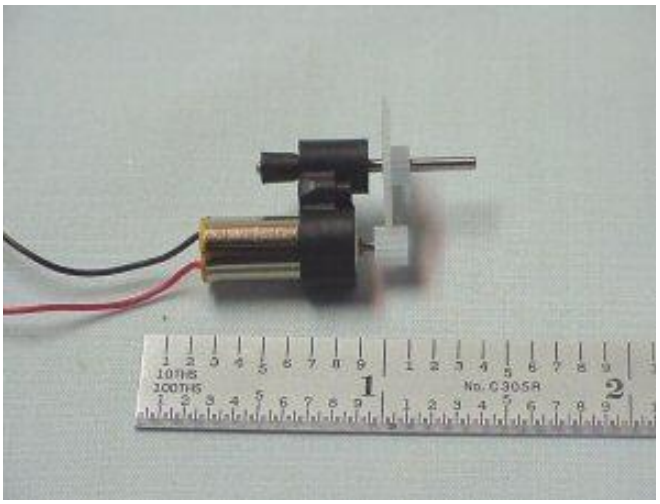


Figure 9. Representation of SS-1.7, SS-2.3, and SS-3.3 Brushed Motors (Only Internal Characteristics Differ) with 4:1 Gear Ratio [13]

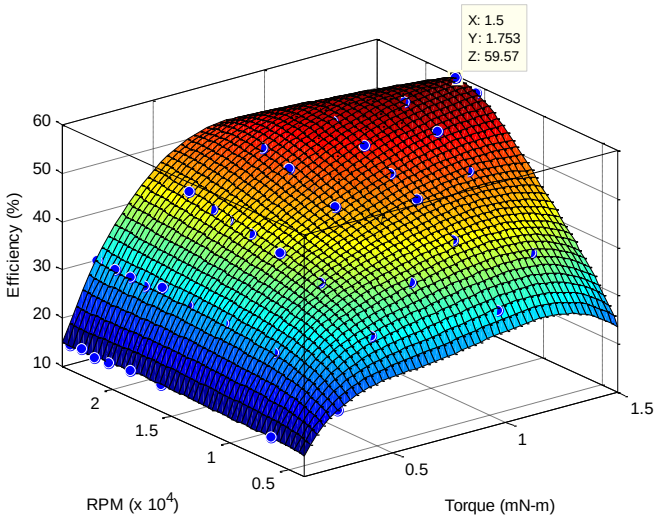


Figure 10. Efficiency as a Function of Torque and RPM for an AP03-4000 Brushless Motor [12]

As stated previously, brushless motors require an electronic speed controller (ESC) to alternatively phase the voltage and current to the motor windings to rotate the motor shaft. This electronic phase shifting is a complicated process that incurs its own power usage which results in a lower effective efficiency for brushless motors [12]. This effect is evident in Figure 11 in which the speed controller efficiency drops as the torque is increased. This results in a decrease in maximum efficiency between 10-20% for brushless motor systems, essentially negating any benefits over brushed motors.

Unlike brushless motors, brushed motors can vary rotational speed simply by changing the supplied voltage. However, in a practical MAV application where a battery is used, the voltage supplied to each motor cannot be directly changed. Instead, brushed ESCs can be used to rapidly switch the current to the motor on and off with a pulse width modulated signal. This effectively decreases the power supplied to the

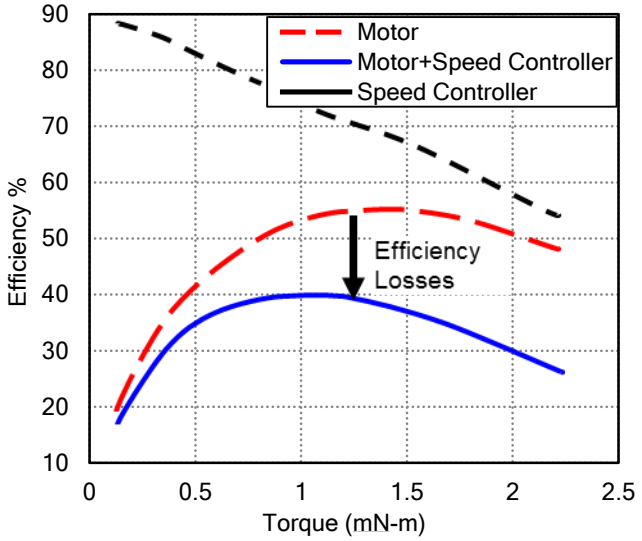


Figure 11. Efficiency Loss for the AP03-7000 Brushless Motor with a Castle Creations Phoenix 6 ESC [12]

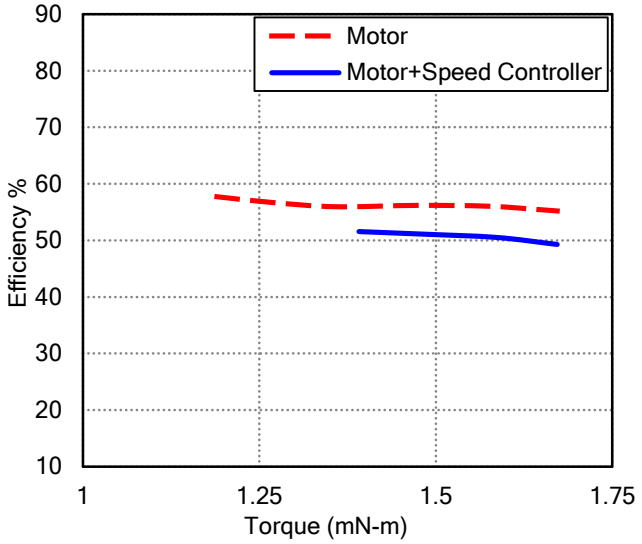


Figure 12. Efficiency Loss for the SS-3.3 Brushed Motor with a HobbyKing 3A Single Cell ESC

motor proportional to the pulse width. This is a much simpler process than compared to the brushless ESC's and does not incur as much power loss. To confirm this, experimental studies were carried out with the SS-3.3 brushed motor and a 3A single cell brushed ESC. These results, shown in Figure 12, were compared with the brushless results from the Harrington, Kroninger study. Though this motor was operating under similar conditions as the AP03-7000, the efficiency loss is much less. In particular, both isolated motors are operating at approximately 55% efficiency at 1.5mN-m of torque. But when ESC power consumption is accounted for, efficiency dropped 18% for the brushless system and only 5% for the brushed system. This same trend was observed with the AP03-4000, SS-1.7 and SS-2.3 motors as well.

Since it is known that maximum motor efficiency is dependent on specific torque and RPM combinations [12], a transmission can be implemented to achieve the optimal combination. This can be a simple two-gear transmission between the motor shaft and rotor. In a direct drive application (i.e. gear ratio 1:1) the RPM and torque of the rotor is same as the motor's. But with a higher gear ratio (i.e. 4:1), the rotor will operate at a higher torque but lower RPM compared to the motor. Ideally, the optimal gear ratio will allow the motor to operate at its maximum efficiency point while rotor still provides the required thrust.

A gearbox will not substantially increase brushless motor performance given the inversely proportional relationship between torque and RPM depending on the gear ratio. It is known from Figure 10 that brushless motors require both high RPM and torque for maximum efficiency. This is best achieved with a direct drive motor-rotor coupling since gear ratios would trade-off torque for RPM or vice versa. However, it was shown in Figure 7 that brushed motors operate most efficiently at low torque and high RPM.

Tests were conducted to determine if geared, brushed motors confirmed increased performance over direct drive brushless motors. SS-1.7, SS-2.3, and SS-3.3 motors were tested with gear ratios between 2:1 and 7:1 and compared against the AP03-4000 brushless motor. The optimized rotor was used in each test to produce 10-20 g of thrust (~1/4 the thrust needed for a 40-80g quadrotor). As an example, the results from the SS-3.3 brushed motor with 4:1 gear ratio are compared with the AP03-4000 in Figure 13. This confirms that the geared brushed motor can produce the same amount of thrust much more efficiently than the brushless motor.

The findings from Harrington and Kroninger have been utilized as a base for further motor optimization studies. It was known that the AP03-4000, AP03-7000, SS-1.7, SS-2.3, and SS-3.3 motors represent the best performance characteristics of small brushless and brushed motors [12]. It was also known that brushed motors operate most efficiently at low torque at high RPM while brushless motors operate

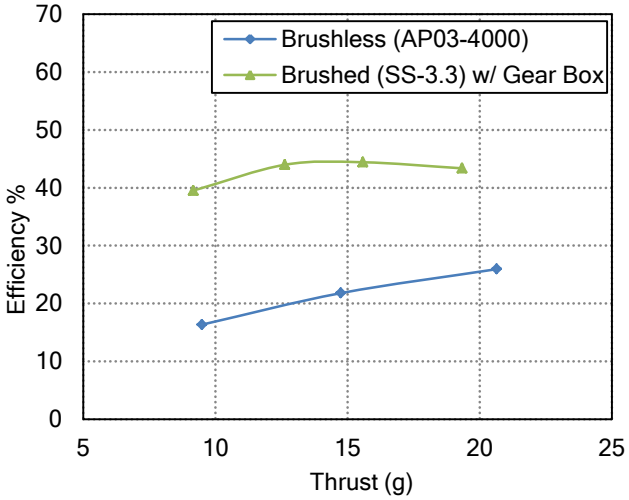


Figure 13. Efficiency Comparison Between a Direct Drive Brushless Motor and a Brushed Motor with 4:1 Gear Ratio

most efficiently at high torque and high RPM [12]. Based on the present experimental studies it was determined that ESCs have a more pronounced effect on power losses for the brushless motors than the brushed motors. It was also determined that geared transmissions are able to provide substantial performance boosts for brushed motor systems. For these reasons, only the SS-1.7, SS-2.3, and SS-3.3 brushed motors were considered in more extensive optimization studies.

Coupled Motor Rotor Experiments

Maximizing the flight endurance of a micro quadrotor requires optimizing the coupling of the optimized rotor with the brushed motor through a power transmission system. While the SS brushed motors were shown to have the highest efficiency with relatively little ESC losses, it was still unknown how this would translate into quadrotor flight endurance. To determine this, two important performance metrics were established: (1) electric power loading (EPL) and (2) and thrust available at an applied voltage (thrust/volt). EPL is another representation for efficiency of the power system. It is simply the thrust produced by the rotor over the electrical power consumption needed to generate that thrust. A higher EPL indicates a higher efficiency of the propulsion system. However, the most efficient systems may not be able to generate the necessary thrust for flight. This is why the second metric was established. A successful, hovering quadrotor must be able to maintain a minimum thrust/volt depending on its gross weight and available battery voltage. Since it was determined that SS brushed motors would be used in the vehicle design, their weights (approximately 2.75g) were factored into the gross weight for a more accurate estimation of 48g. This requires that each motor-rotor pair can supply a minimum of 12g of thrust with the available battery voltage (typically 3.6 – 4.2V).

To determine the highest performance power transmission system, the optimized rotor was paired with each SS brushed

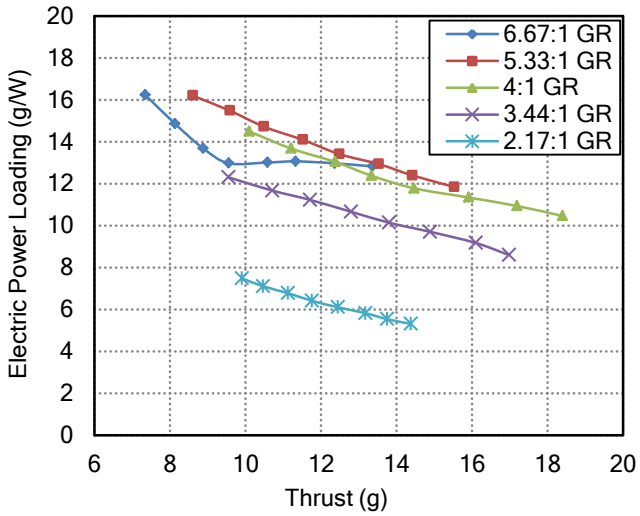


Figure 14. SS-3.33 Motor; Electric Power Loading vs. Thrust with Varying Gear Ratio

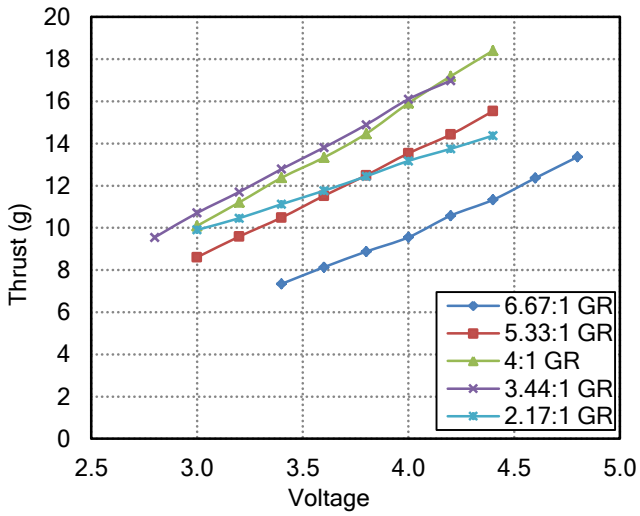


Figure 15. SS-3.33 Motor; Thrust vs. Voltage with Varying Gear Ratio

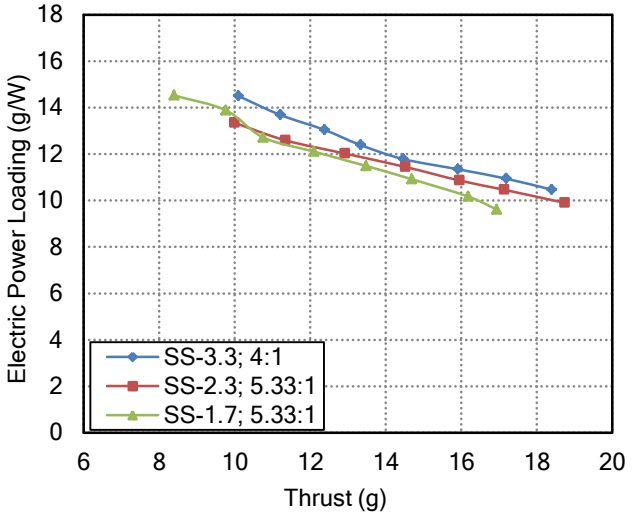


Figure 16. Electric Power Loading vs. Thrust for Optimal Motor and Gear Ratio Combinations

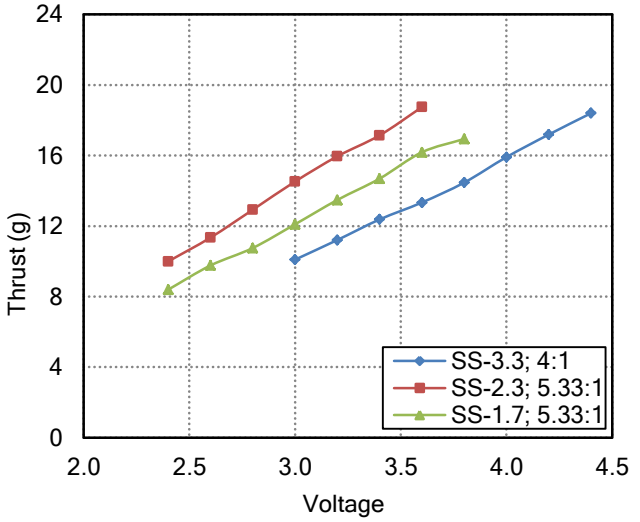


Figure 17. Thrust vs. Voltage for Optimal Motor and Gear Ratio Combinations

motor via gear ratios (GR) ranging from 2:1 to 7:1, for a total of 21 tested transmission configurations. Each configuration was fixed to a thrust measurement stand, and the power available was controlled through a power supply. Each configuration was tested over a voltage range which provided 8 – 18g of thrust since this would be the typical operational range for a micro quadrotor. A summary of results from power transmission configurations are displayed in Figure 14 - Figure 19 in terms of EPL and thrust/volt.

The EPL as a function of thrust for the SS-3.3 motor is displayed in Figure 14. In this chart it is evident that, in general, EPL increases as the gear ratio increases for a given thrust. However, beyond a gear ratio of 5.33:1 the EPL starts dropping. From this perspective, it appears that a gear ratio of 5.33 would yield the highest vehicle flight endurance. However, increasing the gear ratio also has the adverse effect of decreasing the RPM of the rotor relative to the motor shaft. This results in a decrease in thrust for the same

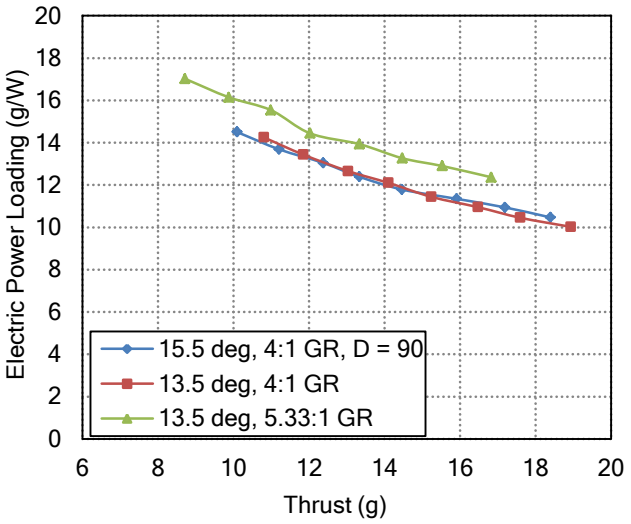


Figure 18. SS-3.3 Motor; Electric Power Loading vs. Voltage for Multiple Pitch and Gear Ratio Combinations

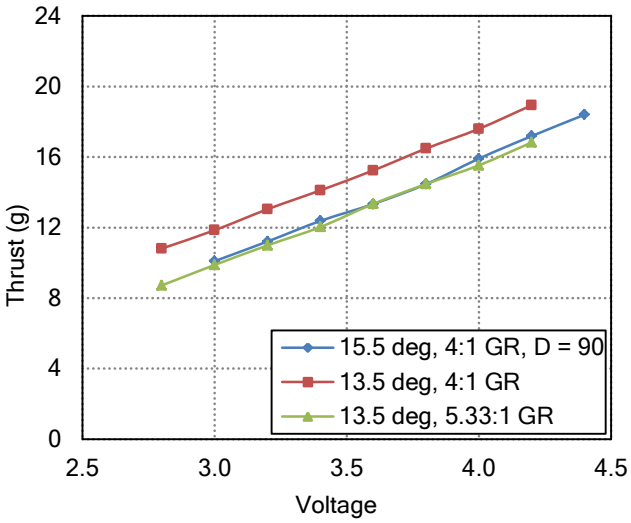


Figure 19. SS-3.3 Motor; Thrust vs. Voltage for Multiple Pitch and Gear Ratio Combinations (110mm Dia.)

voltage, as seen in Figure 15. It is evident in this plot that the 5.33:1 gear ratio may not be able to supply the necessary thrust (12g) at low voltages despite being the most efficient.

Similar tests completed with SS-2.3 and SS-1.7 motors yielded the same trends as the SS-3.3 tests. Again, higher gear ratios tend to increase EPL while decreasing thrust/volt. However, the major difference with these two motors is that they can achieve the same thrust levels as the SS-3.3 but at much lower voltages (2 – 3.5V). Therefore, with a typical battery supplying 3.6 – 4.2V (detailed in the next section), these motors will be able to supply more thrust compared to the SS-3.3. However, it was also observed that they will be less efficient at higher voltages.

To better compare the overall propulsion system characteristics relative to each other, the best performing gear ratio for each motor was obtained. These were the combinations which provided the highest EPL while maintaining more than 12g of thrust for 3.6 – 4.2V. The EPL and thrust of these optimal combinations are re-plotted on Figure 16 and Figure 17 respectively. With these charts, it is more evident that the SS-3.3 is the most efficient while still providing adequate thrust for a given voltage. The efficiency loss from the SS-1.7 and SS-2.3 motors outweighs the benefit from the large excess thrust they can provide. For this reason, the SS-3.3 motor with a 4:1 gear ratio was chosen for the preliminary vehicle design iteration.

Initial flight tests of the preliminary vehicle design iteration showed that the quadrotor would quickly reach a point where no more excess thrust could be provided. Thus, controllability was difficulty to maintain with little excess power. This was due to battery voltage drop off, which will be investigated in the next section. Rather than compromising overall flight endurance by using SS-1.7 and SS-2.3 motors for excess thrust, the optimized rotor diameter was increased from 90mm to 110mm for improved power loading. The final power transmission optimization tests were completed with the larger diameter rotor. This involved iteratively varying the collective pitch of the rotor blades between 11.5^{o} and 17.5^{o} and systematically testing each pitch with the 4:1 and 5.33:1 gear ratios. Two highest performance results from these experiments relative to the previous optimized transmission (15.5^{o} collective, 4:1 GR, dia. = 90mm) are shown in Figure 17 and Figure 18. It was generally observed from these tests that the lower collective pitches increased EPL while decreasing thrust/volt. In particular, it was concluded from these experiments that new power transmission should utilize 13.5^{o} collective pitch in the optimized rotor, a SS-3.3 brushed motor, and a 5.33:1 gear ratio to improve EPL without compromising thrust/volt, or 4:1 gear ratio to improve excess thrust/volt without compromising EPL.

Battery Discharge Tests

To keep the mass of the micro quadrotor below 50g, a lightweight, high energy density power source is required.

While many types of batteries exist, including Nickel Cadmium, Nickel-Metal Hydride, and Lead Acid, the heavy metals used in these batteries contribute too much weight to be feasibly incorporated in the current quadrotor design [14]. Another battery type known as Lithium Ion Polymer (Li-poly) utilizes lightweight Lithium and polymers to achieve a large electrochemical potential, and therefore some of the best energy densities. Li-poly batteries can have 2-3 times the energy density of standard, heavy metal batteries. Therefore, only Li-poly batteries were examined for use on the micro quadrotor.

As stated in the previous section, batteries do not maintain a constant voltage as they are discharged. Rather, the battery voltage drops steadily as the available charge is drained till it reaches an effective cutoff point. For 3.7V rated Li-poly batteries, this is approximately 3.6V. This battery discharge characteristic was studied with the use of a Computerized Battery Analyzer (CBA). Batteries were tested by discharging at a constant current draw and recording the battery voltage over time with the CBA. Results from the power transmission studies showed each motor would draw approximately 0.3 A, so it was initially estimated that the quadrotor would draw approximately 1.2 Amps. Therefore, this was the current draw used in the initial battery tests. Batteries of various capacities, manufacturers, and geometries examined to determine which had the best overall characteristics. Batteries are designated by [manufacturer initial]_[capacity in milliamp-hours]_[geometry: Rectangular or Cylindrical]. The results of the tests are shown in Figure 20 - Figure 22.

It is immediately evident in Figure 20 that battery endurance is most directly affected by the rated capacity. This is to be expected as a larger capacity battery will be able to supply the same power for a longer period of time. It was also noticed that the batteries may not maintain the same voltage levels relative to each at the same % capacity remaining. To make this clearer, the data from Figure 20 was normalized by each battery's final time to discharge completely to obtain the voltage level as a function of normalized time. This normalized data is displayed in Figure 21. This chart shows that the voltage characteristics are heavily dependent on battery geometry. Though each battery has the same rated voltage (3.7V), it is evident in Figure 21 that the two high capacity rectangular batteries (T_950mAh_R and T_750mAh_R) maintain a 1 – 0.5V higher voltage over the two cylindrical batteries (T_9mAh_C and T_650mAh_C). From a controls perspective, the rectangular batteries may be preferred, as the increased voltage means the rotors would be able to supply more excess thrust. The excess thrust would enhance flight speed and maneuverability, as faster roll, pitch, and yaw rates could be achieved with more thrust input to the rotors.

However, excess voltage is not as beneficial from a flight endurance perspective. In the hover scenario, the large excess voltage at the beginning of discharge will not be utilized. The speed controllers will regulate the voltage to each rotor via a pulse width modulation signal to supply only the power necessary for hover. Hence, only the minimal voltage needed for sustained hover thrust is required from the battery. The results from the voltage studies in Figure 20, confirm that the lowest effective voltage supplied is approximately the same for all for all batteries (3.6V). Thus, the weight of the battery becomes a bigger factor than lowest effective voltage for hover endurance. As the battery capacity increases so does the weight. Thus, the higher capacity batteries will also require a larger current draw and may not actually produce the largest endurance. A more direct relation between battery weight and endurance was required.

Determining an estimated current draw for each battery required an estimation of the final quadrotor weight. Since the rotors, gears, motors, ESCs, and processor-sensor board to be used were already known from previous studies, an accurate empty weight for the vehicle was estimated at

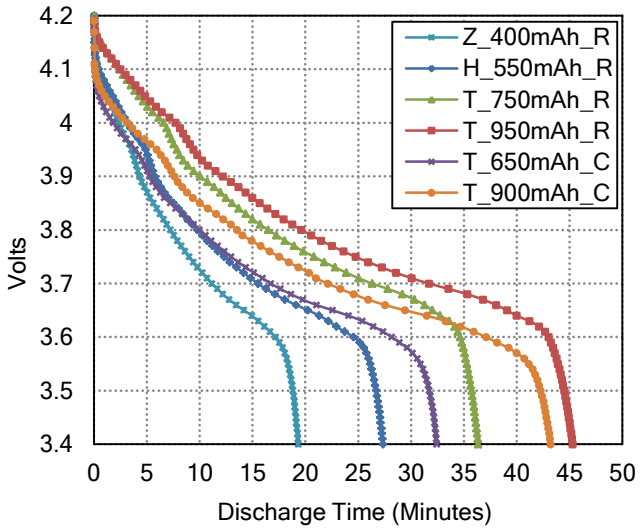


Figure 20. Battery Voltage Available Over Time under 1.2 Amp Current Draw

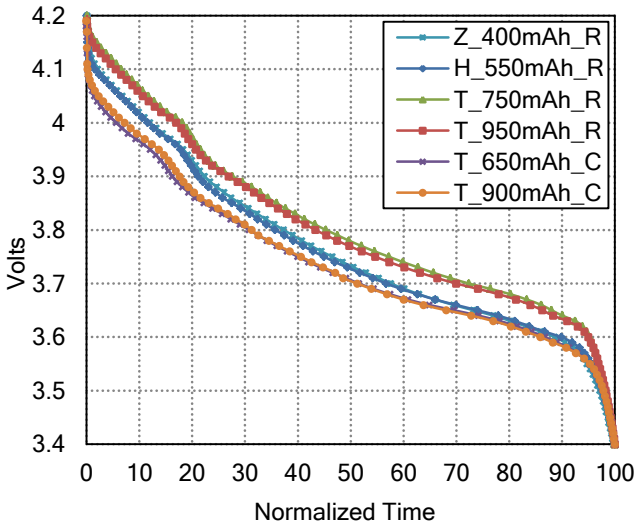


Figure 21. Battery Voltage Available as a Function of Normalized Time

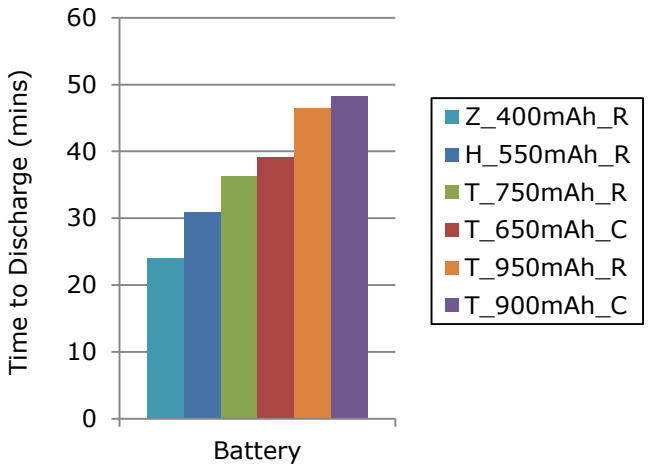


Figure 22. Each Battery's Time to Completely Discharge Under Individual Current Draw

31.5g. By adding the individual battery weight, and dividing by 4, an accurate estimation of individual rotor thrust could be obtained. By interpolating measured current and thrust data from the power transmission studies, an approximate value of total current draw in hover could be obtained. Each battery was then discharged at its specific current draw for the most accurate prediction of hover endurance. The results of this study, displayed in Figure 22, indicate battery endurance is dependent on rated capacity as well as battery geometry. Specifically, the two cylindrical batteries have a longer discharge time for their capacity as opposed to the rectangular batteries. This could be due to a more efficient use of cathode/anode surface area within the volume of the battery. Though, more in-depth analysis is required to determine exactly why this is the case. Though the T_950mAh_R battery showed the second highest predicted endurance, it was rejected as a viable option since it caused the quadrotor to weigh more than the desired 50g. Ultimately, only the two cylindrical batteries were chosen for further testing in the full vehicle design. The T_650mAh_C weighs 12.75g and the T_900mAh_C weighs 17g.

Avionics and Telemetry

To control and stabilize the quadrotor during flight, a high-bandwidth attitude feedback control is required. This was made possible by incorporating a lightweight (2 g) processor-sensor board (GINA2.2 developed by U.C.Berkeley [15]) on the vehicle. The principal components of this board are a TI MSP430 microprocessor for onboard computation tasks, ITG3200 tri-axial gyros, KXSD9 tri-axial accelerometer, and an Atmel radio and antenna for wireless communication tasks. The wireless communication has a latency less than 20–30 ms. The time-critical inner-loop feedback occurs at an update rate of 3 ms [16]. The user communicates with the vehicle using a LabVIEW interface. The feedback control system was implemented using a telemetry setup. A 2.4 GHz Atmel AVR transceiver was attached to a base station. This was used to wirelessly update (IEEE 802.15.4 protocol) the feedback gains, trim inputs, and pilot commands to the vehicle in flight [16].

Inner-Loop Feedback Control System

The gyros on the board measure the pitch q, roll p, and yaw r attitude rates, while the accelerometers record the tilt of the gravity vector. The vehicle attitude can be extracted by integrating the gyro measurements with time. However, it is known that this leads to drift in attitude measurements [17]. Accelerometers, on the other hand, offer stable bias but are sensitive to vibrations and, in general, offer poor high-frequency information [18]. Therefore, a complementary filter was incorporated to extract the pitch and roll Euler angles using a high-pass filter for the gyros (4 Hz cutoff) and a low-pass filter for accelerometers (6 Hz cutoff). The onboard inner-loop feedback was implemented using a proportional–derivative (PD) controller as shown in Figure 23. The feedback states were the pitch and roll Euler angles (θ, ϕ) and the attitude rates (p, q, and r). An outer-loop feedback capability was provided for vehicle control by a human pilot. The final control inputs to the vehicle actuators

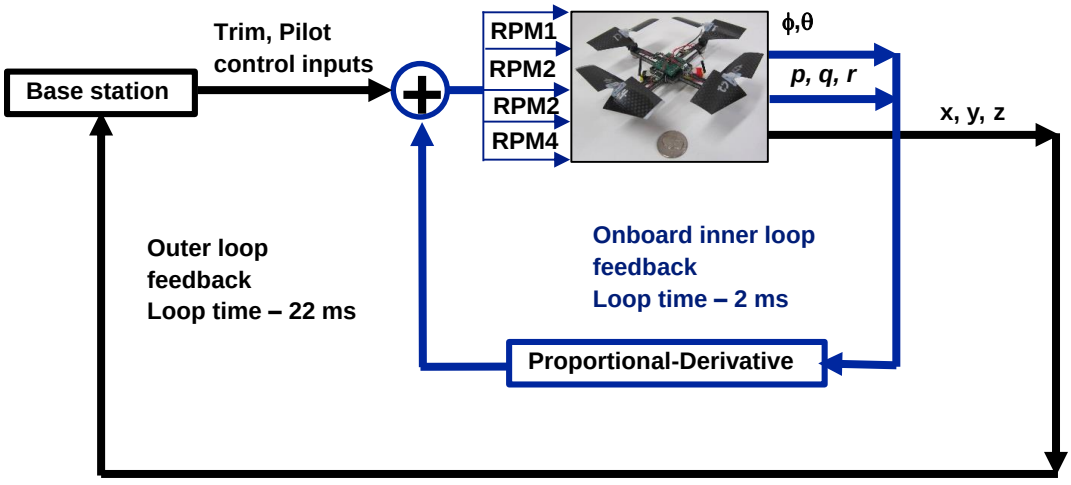


Figure 23. Schematic of the Implemented Feedback Control System

are the individual rotational speeds for the four rotors Figure 23.

Airframe Fabrication and General Configuration

The final component needed to construct the optimized quadrotor was an ultra-lightweight, structural airframe to hold the other components (motors, battery, and processor-sensor board). It was crucial that the airframe minimized the gross weight of the vehicle while still maintaining durability. For this reason, the airframe was chosen to be milled out of a high strength-to-weight carbon fiber-balsa wood composite sheet. With 1.5mm thick connecting spars, the airframe weight is only 3.3g. The general quadrotor configuration with all components integrated into the airframe is shown in Figure 24. To understand the significance of the airframe weight reduction, the weights of each component were compared with a representative sample of similarly sized quadrotors. As seen in Figure 25, the airframe weight of the optimized quadrotor takes up 23 - 43% less of the total vehicle weight compared to other models.

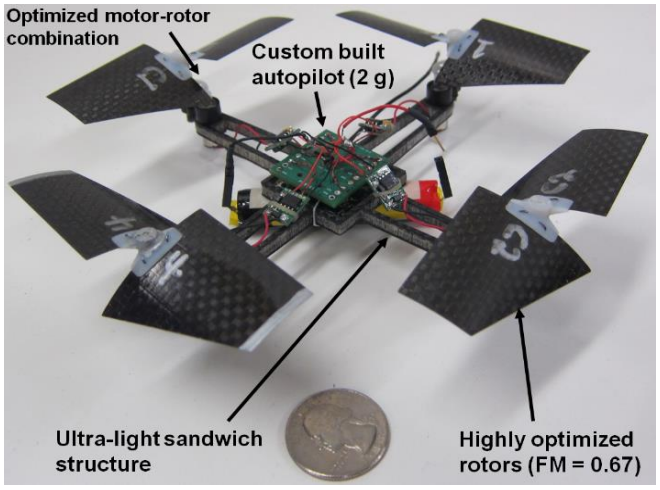


Figure 24. General Micro-Quadrotor Configuration

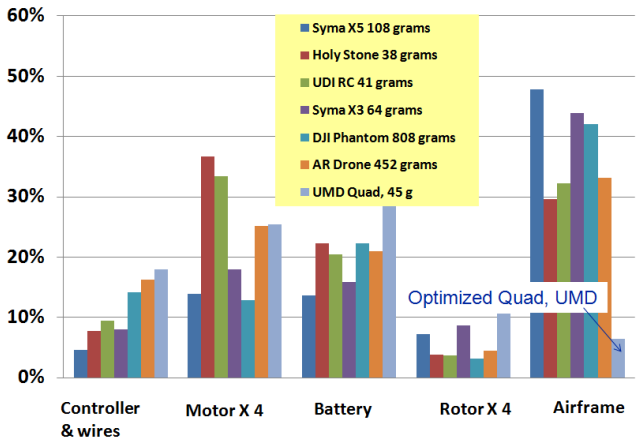


Figure 25. Comparison of the Optimized Quadrotor Component Weight % with a Representative Sample of Commercial Quadrotors

Gimbal Testing

Before flight testing the quadrotor for endurance in free hover, gimbal tests were conducted with the use of the setup pictured in Figure 26. An isolation mount supported the quadrotor on a thrust measurement balance. The quadrotor was connected to the mount via a gimbal bearing. The gimbal setup mimics free hover since it allows the quadrotor to yaw, pitch, and roll freely, but constrains it in translational motion. In this way, the quadrotor must expend energy to offset its weight as well as to stabilize itself as it would in a hover scenario. The quadrotor thrust was monitored on the balance to determine when it could no longer offset its own weight on a single battery charge. The purpose of the gimbal tests was to determine which final configuration would be suitable for free hover testing. Two optimal gear ratios and two high energy density batteries were tested with each other to determine which configuration should yield the highest endurance. The results of the experiment are shown in

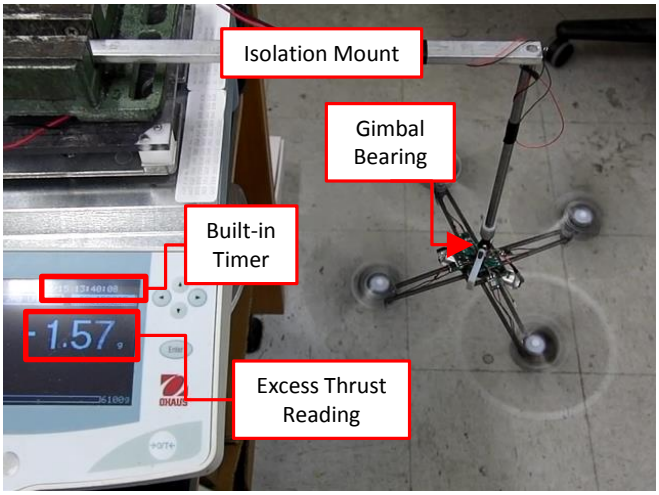


Figure 26. Gimbal Testing Setup

Table 2. Gimbal Test Endurance Times

Gear Ratio	Collective Pitch	Battery	Gross Weight (g)	Endurance (minutes)
4:1	13.5°	650 mAh	44.44	32
4:1	13.5°	900 mAh	48.8	37
5.33:1	13.5°	650 mAh	44.13	40.5
5.33:1	13.5°	900 mAh	48.5	49



Figure 27. Optimized Quadrotor in Free Hover

Table **2**. These results show that for either gear ratio, the T_900mAh_C battery will yield a higher predicted endurance, and for either battery, the 5.33:1 gear ratio will yield a higher endurance. This is likely due to the higher energy density of the 900mAh battery and the higher EPL of the 5.33:1 gear ratio. Furthermore, these tests validated the battery discharge tests since the two endurance values for the 5.33:1 GR match the corresponding battery discharge times within +/- 2 minutes.

Free Flight Hover Testing

The final quadrotor configuration used the SS-3.3 motor with a 5.33:1 gear ratio since this was determined to yield the highest predicted endurance by the gimbal tests. Both the T_650mAh_C and T_900mAh_R battery were tested. Free hover tests were conducted by a human pilot within a 20 ft^3 testing area pictured in Figure 27. Hover endurance was measured from the time the quadrotor took off to the time it could no longer stabilize itself and support its own weight in one continuous flight. While the gimbal tests predicted that the 900mAh battery would provide the largest endurance, this was difficult to confirm in free hover since this quadrotor configuration became less controllable as the battery discharged. It is likely the added weight changes the dynamics of the system in a way that must be accounted for in the control algorithm. Flight tests for improved continuous hover endurance with the 900mAh battery are a planned area of research.

When equipped with the T_650mAh_C battery, the quadrotor maintained controlled hover 31 minutes. However, this has the potential to be improved, as it is still below the endurance predicted by the gimbal and battery discharge tests. It was observed in the flight tests that the control authority and stability characteristics diminish as the battery is discharged. This indicates that the magnitude of control inputs varies significantly as the battery is discharging. To mitigate this, the differential gains in the PD feedback control loop were increased to be nominally higher than gains required for hover flight under a full battery charge. Future work will include an adaptive gain scheme such that the flight characteristics remain the same throughout the duration of battery discharge, thus improving controllability and reducing pilot workload. The present weight breakdown of the optimized quadrotor design is shown in Table 3.

Table 3. Micro-Quadrotor Weight Breakdown

Component	Weight (g)	% Weight
Controllers and Wires	8.51	19.12
SS-3.3 Motors	11	24.72
650mAh Battery	12.73	28.61
Rotors	4.6	10.34
Gear Transmissions	4.36	9.80
Airframe	3.3	7.42
Total	44.5	

Conclusions

The present research has focused on designing and optimizing a MAV-scale quadrotor helicopter for maximum hover endurance. This has been achieved through systematic testing of each component, including the motors, gear transmission systems, and battery, as well as significantly reducing the airframe weight of the quadrotor. Characteristics of an aerodynamically optimized micro-rotor from a previous study were applied to a new 110mm diameter rotor used in the quad rotor design. The optimal rotor was fabricated from carbon fiber composite and has the following parameters: 2 blades, 0.32 thrust weighted solidity, 21.4mm chord at 75% span, 0.25mm thickness, 5-10% camber, 13.5° of collective blade pitch, -11.4° of twist, and a 0.5 chord taper ratio. Optimization studies of the other subsystems yielded the following results:

1. Brushed motors are more suitable than brushless motors for use on an MAV-scale quadrotor. Brushless motors and electronic speed controllers (ESCs) are heavier than brushed motors and brushed ESCs. It was determined that ESCs have a much more pronounced decrease in efficiency in brushless motor systems than in brushed motor systems. It was also determined that geared transmissions are able to provide substantial performance boosts for brushed motor systems but only marginally effect brushless motors. For these reasons,

the three most efficient brushed motors were considered in more extensive optimization studies.

2. Identically sized brushed motors with different internal resistances will exhibit different power loading and thrust/volt characteristics. High resistance brushed motors (3.3 Ohms) exhibit higher electrical power loading and lower thrust/volt compared to lower resistance motors (1.7 and 2.3 Ohms).

3. Pairing brushed motors with larger gear ratios has the effect of improving electric power loading while reducing thrust/volt. Smaller gear ratios decrease EPL while increasing thrust/volt. For the same gear ratio, high resistance brushed motors had higher EPL than low resistance motors but produced less thrust/volt. Brushed motors with 4:1 and 5.33:1 gear ratios had the highest overall efficiency while still being able to supply the minimum thrust/volt for flight.

4. Li-poly Battery voltage does not remain constant as it is discharged. The amount of voltage drop is dependent on battery geometry. Rectangular batteries show less voltage drop than cylindrical batteries. The lowest sustainable voltage for 3.7V rated Li-poly batteries is effectively 3.6V. Energy density is also dependent on battery geometry. Cylindrical batteries have longer endurance for their mass than rectangular batteries at the required current draw. Due to this characteristic, the 650mAh and 900mAh cylindrical batteries were studied in the full vehicle configuration.

5. Vehicle control and stability were provided by a lightweight (2g) processor-sensor board (GINA-Mote). A telemetry setup with a base station was used to wirelessly update the feedback gains, trim inputs, and pilot commands to the vehicle in flight.

6. The quadrotor airframe was fabricated from an ultra-lightweight carbon fiber-balsa wood composite for significant weight reduction. This comprises only 7.4% of the optimized quadrotor gross weight compared to 30 - 50% for other similarly sized quadrotors.

7. The highest predicted endurance times on the gimbal test setup were achieved with the 5.33:1 gear ratio. The 650mAh and 900mAh batteries resulted in 40.5 minutes 49 minutes of predicted endurance, respectively. In free hover flight tests, the maximum endurance achieved with the 650mAh battery was 31 minutes.

8. Future work will focus on optimizing the control system. An adaptive gain scheme will be implemented such that the flight characteristics remain the same throughout the duration of battery discharge to improve controlled flight endurance and reduce pilot workload.

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